Socio3: A Hybrid Blockchain-IPFS Architecture for Decentralized Social Media with On-Chain Content Moderation

Tejas Lalit Patil*, Atharva Anant Baodhankar*, Manas Amol Prabhu*, and Chaitanya Bhosale†
*Department of Computer Engineering,
MIT Academy of Engineering, Alandi, Pune, Maharashtra, India
Email: {tejaspatil1936, baodhankaratharva, manas28prabhu}@gmail.com
†Assistant Professor, Department of Computer Engineering,
MIT Academy of Engineering, Alandi, Pune, Maharashtra, India

***Abstract*—Conventional social media platforms concentrate content ownership, moderation authority, and revenue distribution within a single corporate entity, exposing users to data exploitation, opaque censorship, and platform lock-in. This paper presents Socio3, a production-grade decentralized social media application that distributes these responsibilities across Ethereum smart contracts, the InterPlanetary File System (IPFS), and a selective Firebase layer. The platform employs three modular Solidity contracts—PostContract, SocialContract, and ProfileContract—deployed on the Sepolia testnet to manage content publication, social-graph interactions, and wallet-based identity, respectively. Media assets are stored on IPFS through Pinata, while content hashes are anchored on-chain to guarantee immutability and verifiable ownership. A dual-path content moderation mechanism combines community-driven reporting with algorithmic auto-removal governed by configurable report thresholds and a report-to-like ratio, with every moderation action recorded as a transparent blockchain event. Direct creator monetization is achieved through a zero-fee Ether tipping system that transfers funds atomically via smart contract logic. Functional evaluation across twelve core operations confirms correct execution, with gas costs ranging from 45 000 to 200 000 units per transaction. The hybrid architecture reduces on-chain storage costs by over 99% compared to fully on-chain alternatives while preserving sub-second latency for real-time features. These results demonstrate that thoughtful partitioning of on-chain and off-chain responsibilities can yield a decentralized social platform whose usability approaches that of centralized incumbents.**

***Index Terms*—Blockchain, Decentralized Social Media, Smart Contracts, IPFS, Content Moderation, Ethereum**

I. Introduction

Social media platforms now mediate a significant share of global communication, commerce, and cultural production. Platforms such as Instagram, X (formerly Twitter), and Facebook collectively serve billions of users, yet the architectural foundations of these services remain firmly centralized. A single corporate entity stores user-generated content, curates algorithmic feeds, adjudicates moderation disputes, and captures the overwhelming majority of advertising revenue generated from user activity. The consequences of this concentration have become increasingly visible: large-scale data breaches expose private information, opaque recommendation algorithms amplify polarizing content, and creators receive only a fraction of the economic value their work generates [1], [2].

The problem extends beyond privacy. Platform lock-in prevents users from migrating their follower networks and content archives to competing services, effectively binding them to a single provider's terms of service. Content removal decisions—whether motivated by policy enforcement, government pressure, or commercial interest—occur without public accountability, and affected users have limited recourse [2]. Meanwhile, the creator economy remains constrained by intermediary fees that range from 30% to 50% of creator earnings on dominant platforms [3]. The decentralized social media (DSM) market is projected to grow from $18.5 billion in 2025 to over $141 billion by 2035 [4], signaling substantial industry recognition that centralized models are approaching structural limits.

Blockchain technology offers a structural alternative. By recording state transitions on a distributed, append-only ledger, blockchain systems can guarantee data integrity, enable transparent governance, and support peer-to-peer value transfer without trusted intermediaries [5]. Several DSM platforms have emerged to exploit these properties. Lens Protocol treats the social graph as an on-chain infrastructure of Non-Fungible Tokens (NFTs) [4], [6]. Farcaster separates identity registration on Ethereum from data propagation across a network of peer-to-peer hubs [4]. Steemit and DeSo implement native social-ledger designs with token-based incentive mechanisms [1]. Each of these platforms, however, confronts a recurring tension between decentralization and user experience. Fully on-chain architectures impose prohibitive gas costs and high latency on routine social interactions, while protocol-level solutions often demand considerable technical sophistication from end users [4], [7].

The motivation for Socio3 arises from this gap. Rather than forcing all data onto the blockchain or delegating all storage to centralized databases, Socio3 partitions responsibilities according to the value and permanence of each data type. Content ownership records, social-graph relationships, financial transactions, and moderation audit trails reside on

Ethereum smart contracts, where immutability and transparency are essential. Media files are stored on IPFS, exploiting content-addressed hashing for integrity verification at a fraction of on-chain storage costs. Ephemeral, high-frequency data—comments, notifications, and presence indicators—flow through Firebase Firestore, which provides real-time synchronization without gas expenditure. The result is a platform that preserves blockchain guarantees for critical operations while delivering the responsiveness users expect from conventional social applications.

The principal objectives of this work are as follows:

1) Design and implement a modular smart contract architecture that separates post management, social interactions, and profile identity into independently deployable and auditable contracts.
2) Integrate IPFS-based decentralized storage with on-chain metadata anchoring to eliminate centralized content servers while keeping transaction costs practical.
3) Develop a dual-path content moderation system that combines community-driven reporting with algorithmic auto-removal, recording all actions transparently on-chain.
4) Enable direct, zero-fee creator monetization through an atomic Ether tipping mechanism embedded in the social contract.
5) Evaluate the platform's gas efficiency, functional correctness, and architectural trade-offs against existing DSM solutions.

The specific contributions of this paper include:

- A **hybrid four-layer architecture** that strategically distributes data across Ethereum, IPFS, and Firebase based on permanence, value, and access-frequency characteristics.
- A **modular three-contract design** (PostContract, SocialContract, ProfileContract) that improves gas efficiency, isolates security surfaces, and supports independent upgradability.
- An **on-chain moderation algorithm** with configurable thresholds—including absolute report count, zero-engagement removal, and report-to-like ratio criteria—that executes deterministically within the smart contract.
- A **frictionless onboarding mechanism** combining wallet-based authentication with an automated testnet Ether gifting system.
- A **deployed, functional platform** accessible at `socio3.vercel.app` with comprehensive **functional and gas-cost evaluation** on the Sepolia testnet.

The remainder of this paper is organized as follows. Section II surveys related work on DSM platforms, blockchain-based moderation, and creator monetization. Section III describes the proposed system architecture, smart contract design, and moderation algorithm in detail. Section IV presents implementation specifics. Section V reports functional test results, gas consumption, and comparative analysis. Section VI concludes the paper and outlines future directions.

II. Related Work

Research on blockchain-based social media spans platform architecture, content moderation governance, economic incentive design, and privacy-preserving identity management. This section organizes the existing literature along these dimensions and establishes the research gap that Socio3 addresses.

A. Decentralized Social Media Architectures

Guidi et al. [1] provided an early taxonomy of decentralized social networking approaches, distinguishing between fully on-chain systems, federated architectures, and hybrid designs. Their analysis highlighted the fundamental trade-off: storing all social data on a blockchain ensures censorship resistance but imposes storage costs and latency that render the platform impractical at scale. Acker and Murthy [2] extended this analysis through a scoping review of 134 publications, finding that roughly 75% of existing research focuses on technological implementation—consensus throughput, encryption, storage optimization—while neglecting socio-technical dimensions such as user behavior under token incentives.

Lens Protocol pioneered the concept of the social graph as on-chain infrastructure by representing profiles, follows, and content-collection actions as NFTs on Polygon [4], [6]. Its Momoka Layer-3 solution processes high-frequency transactions off-chain for later settlement, and its 2025 migration to Lens Chain (ZKsync's ZK Stack) further reduced costs [4]. Farcaster adopted a different hybrid strategy: user identities are anchored on Ethereum via Optimism, while social data propagates across a peer-to-peer hub network. The introduction of Snapchain improved scalability by partitioning data by user identifier [8]. DeSo implemented a purpose-built social ledger with native data primitives, trading interoperability for throughput [1]. Mastodon demonstrated the subsidiarity model where individual instances set their own moderation policies [9]. Torres and Mendes [10] proposed a privacy-preserving social network combining on-chain reputation with censorship resistance.

B. Blockchain-Based Content Moderation

Content moderation on decentralized platforms resists the binary choice between unmoderated expression and centralized editorial control. Zia et al. [9] studied moderation across 4 025 Fediverse instances, demonstrating that community-driven standards can coexist within a decentralized topology. Al-Jaroodi and Mohamed [5] concluded that immutable audit trails increase moderation trust but most systems lack configurable, adaptable thresholds. Zafar and Awan [11] explored AI-blockchain integration for detecting cyberbullying. Kenis [12] examined human rights implications of AI-powered moderation. The 2025 Oxford Internet Institute survey [13] found 79% of users support moderation for violence-inciting material, yet opinion divides on responsibility—underscoring the need for auditable mechanisms.

TABLE I
COMPARISON OF DECENTRALIZED SOCIAL MEDIA PLATFORMS

Platform	Storage Model	Moderation	Monetization	Identity	End-User App	Gas Strategy
Lens Protocol	On-chain + L3	Open Actions	NFT-based	Profile NFT	SDK only	ZK-Rollup
Farcaster	P2P Hubs	Community Hubs	Tips + Subs	On-chain DID	Client apps	L2 (Optimism)
DeSo	Full on-chain	Validator vote	Native token	DeSo Identity	Built-in	Custom ledger
Steemit	Full on-chain	DPoS voting	Token rewards	Account-based	Built-in	DPoS
Mastodon	Federated local	Instance-level	Donations	Email/password	Built-in	N/A
Socio3	**IPFS+Chain+FB**	**On-chain algo.**	**ETH tipping**	**Wallet (MetaMask)**	**React SPA**	**3-contract split**

C. Creator Monetization and Tokenomics

Fan et al. [8] analyzed Farcaster's pluralistic economic design, finding that tipping remains asymmetric with a small user fraction generating most transfers. Catalini and Gans [3] observed that platforms forgoing 30–50% commissions achieve better creator retention through token-based alignment. Wang and He [14] described burn-and-mint equilibrium models. Zhang and Li [15] analyzed blockchain tokenomics impact on business models. Gupta and Rai [16] and Chen and Liu [17] examined NFT-based content ownership frameworks.

D. Identity, Privacy, and Security

The Dock Labs framework [18] described Self-Sovereign Identity through DIDs, Verifiable Credentials, and digital wallets. Zhang et al. [19] identified smart contract vulnerabilities, address clustering, and flash-loan attacks as primary threats. Politou et al. [20] noted public blockchain transparency conflicts with GDPR. Finck [21] proposed off-chain personal data storage with on-chain hash anchoring—closely aligned with Socio3's architecture. Batool and Ahmad [22] explored AI-blockchain synergy for governance applications.

E. Research Gap and Positioning

TABLE I summarizes prominent DSM platforms alongside Socio3. Existing solutions tend toward two extremes: fully on-chain designs sacrificing usability (DeSo, Steemit), or protocol-level frameworks requiring developers to build client applications (Lens, Farcaster). None combines modular smart contract moderation with hybrid storage and a production-ready end-user application. Socio3 addresses this gap.

III. PROPOSED SYSTEM ARCHITECTURE

Socio3 is structured as a four-layer architecture where each layer addresses a distinct concern: user interaction, external service integration, on-chain business logic, and persistent data storage.

A. Architectural Overview

Fig. 1 presents the system block diagram. The architecture separates the React frontend from three independent backends: Firebase for real-time data, IPFS (Pinata) for media storage, and Ethereum (MetaMask/Infura) for smart contract execution.

The **Client Layer** consists of a React 19 SPA bundled with Vite 7, styled with Tailwind CSS 4, and animated through Framer Motion 12. MetaMask exposes the EIP-1193 API for account management and transaction signing. Ethers.js v6 abstracts contract calls into typed invocations.

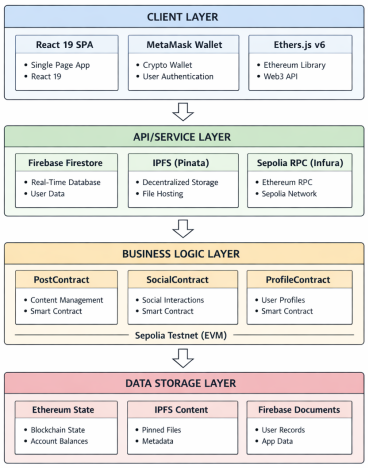


Fig. 1. System block diagram illustrating the four-layer architecture: Client Layer (React 19 SPA, MetaMask, Ethers.js), API/Service Layer (Firebase, IPFS/Pinata, Sepolia RPC), Business Logic Layer (three Solidity contracts), and Data Storage Layer (Ethereum state, IPFS content, Firebase documents).

The **API/Service Layer** bridges the client to three backends. Firebase Firestore provides real-time synchronization for comments, notifications, and presence tracking. Pinata handles IPFS media uploads through authenticated REST endpoints. Infura supplies a Sepolia RPC endpoint for read operations.

The **Business Logic Layer** comprises three Solidity contracts on Sepolia (Section III-C). The **Data Storage Layer** persists state across Ethereum's state trie, IPFS content-addressed storage, and Firestore documents.

B. Use Case Model

Fig. 2 shows three actor types and their system interactions. The *User* performs core social operations: registration, profile management, posting, liking, commenting, following, and searching. The *Content Creator* extends the User with monetization capabilities. The *System Admin* has elevated moderation privileges.

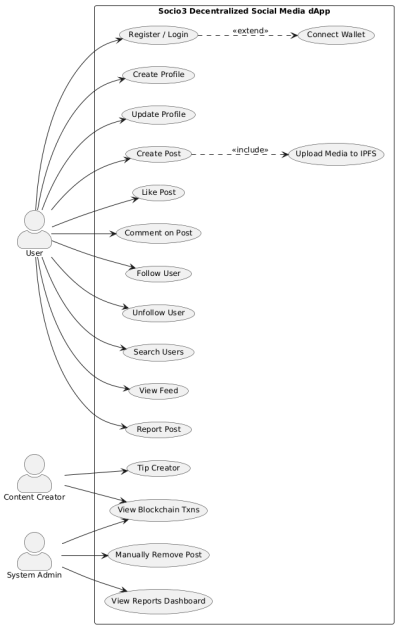


Fig. 2. Use case diagram showing three actors (User, Content Creator, System Admin) and their interactions with the Socio3 dApp.

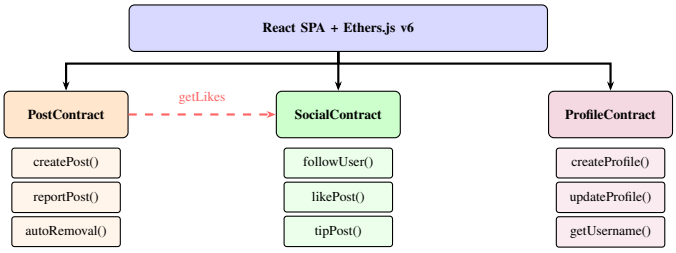


Fig. 3. Smart contract interaction diagram. The frontend communicates independently with each contract via Ethers.js.

C. Modular Smart Contract Design

Separating business logic into three contracts provides: separation of concerns, gas optimization, security isolation, and independent upgradability. Fig. 3 shows how contracts interact with the frontend, each maintaining independent state.

1) PostContract: PostContract manages content lifecycle: creation, retrieval, reporting, and removal. Each post is a `Post` struct with author address, IPFS metadata hash, timestamp, sequential ID, removal flag, and reason string. Listing 1 shows the creation function.

```
function createPost(string memory _ipfsHash)
    public {
    require(bytes(_ipfsHash).length > 0,
        "IPFS hash cannot be empty");
    postCount++;
    posts[postCount] = Post({
        author: msg.sender,
        ipfsHash: _ipfsHash,
        timestamp: block.timestamp,
        id: postCount,
        isRemoved: false,
        removalReason: ""
    });
    emit PostCreated(postCount, msg.sender,
        _ipfsHash, block.timestamp);
}
```

Listing 1. Post creation function in PostContract.sol

2) SocialContract: SocialContract handles following (`following[follower][followed]` mapping with counters), liking (`hasLiked[postId][user]` with `likesCount`), and tipping. The tip function accepts Ether, updates state *before* external transfer (Checks-Effects-Interactions), and emits a `PostTipped` event. Listing 2 shows the implementation.

```
function tipPost(uint256 _postId,
    address _recipient) public payable {
    require(msg.value > 0,
        "Tip must be greater than 0");
    require(_recipient != address(0),
        "Invalid recipient");
    // Effects before interaction
    tips[_postId] += msg.value;
    totalTipsReceived[_recipient] += msg.value;
    // Interaction (external call)
    payable(_recipient).transfer(msg.value);
    emit PostTipped(_postId, msg.sender,
        _recipient, msg.value);
}
```

Listing 2. Tip function with reentrancy protection

3) ProfileContract: ProfileContract manages identity through wallet-to-profile mappings. Profiles store an IPFS hash (pointing to JSON with display name, bio, avatar, cover image), a timestamp, and an existence flag. Bidirectional username mappings enforce uniqueness with 3–20 character validation.

D. Decentralized Storage Integration

Storing a 1 MB image on Ethereum would cost over $200 000 at mainnet gas prices [23]. Socio3 stores media on IPFS through Pinata, recording only the content-addressed hash on-chain. Fig. 4 shows the complete profile creation flow including IPFS upload and blockchain anchoring.

The upload pipeline: (1) client compresses image to $\sim$1 MB via Canvas API at 0.85 quality, (2) uploads to Pinata's API, (3) receives a CID, (4) constructs JSON metadata with CID and caption, (5) uploads metadata to IPFS, (6) passes metadata CID to the smart contract. IPFS hashes are deterministic—any content alteration produces a mismatching hash, rendering tampering detectable [24].

E. Hybrid Data Distribution

TABLE II classifies data by storage destination and rationale, ensuring immutable data resides on-chain, content-addressed media on IPFS, and ephemeral data on Firebase.

F. Content Moderation Algorithm

Socio3 implements dual-path moderation: community reporting with algorithmic auto-removal, and administrator manual removal. Both produce on-chain `PostRemoved` events.

1) Report Submission: Users invoke `reportPost(postId, reportType, reason)` with five categories: Spam (1), Inappropriate (2), Harassment (3), Copyright (4), Other (5). Guards enforce: post existence, no duplicate reports, no self-reporting.

2) Algorithmic Auto-Removal: After each report, `_checkAutoRemoval(postId)` evaluates three criteria. Let r = reports, T = threshold (default 5), l = likes, ρ = ratio (default 2):

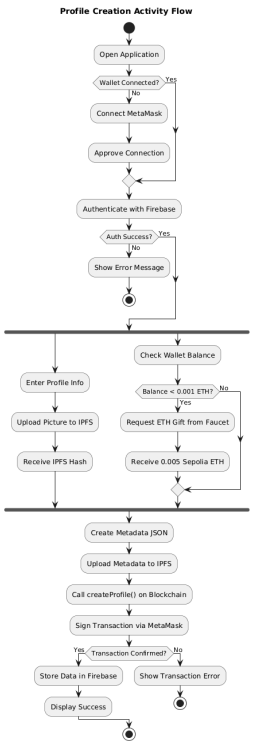


Fig. 4. Activity diagram for profile creation showing wallet connection, Firebase authentication, IPFS media upload, on-chain profile creation with transaction signing, and error handling paths.

TABLE II
DATA DISTRIBUTION ACROSS STORAGE LAYERS

Data Type	Store	Rationale
Post ownership	Ethereum	Immutable provenance
Social graph	Ethereum	Censorship-resistant
Tips/payments	Ethereum	Financial auditability
Report records	Ethereum	Moderation transparency
Profile hashes	Ethereum	Identity anchoring
Media files	IPFS	Cost-efficient, content-addressed
Metadata JSON	IPFS	Paired with media CIDs
Comments	Firebase	Real-time, deletable, no gas
Notifications	Firebase	Ephemeral, instant delivery
Presence status	Firebase	Volatile, high-frequency
ETH gift tracking	Firebase	Anti-fraud control

$$\text{Remove if } r \geq T \quad \text{(absolute threshold)} \tag{1}$$

$$\text{Remove if } r \geq 3 \wedge l = 0 \quad \text{(low engagement)} \tag{2}$$

$$\text{Remove if } r \geq l \times \rho \quad \text{(ratio-based)} \tag{3}$$

Both T and ρ are owner-adjustable, enabling governance-level tuning. Algorithm 1 presents the complete procedure.

Algorithm 1: Dual-Criteria Content Auto-Removal

Input: Post ID p, report count r, threshold T, ratio ρ
Output: Removal decision
if $r \geq T$ **then**
  Mark post p as removed;
  Emit `PostRemoved`(p, author, reason);
  return `true`;
end
if $r \geq 3$ **then**
  $l \leftarrow$ `SocialContract.getLikesCount(`p`);`
  if $l = 0$ **then**
    Mark post p as removed;
    return `true`;
  end
  if $r \geq l \times \rho$ **then**
    Mark post p as removed;
    return `true`;
  end
end
return `false`;

3) Manual Removal: The owner invokes `manuallyRemovePost(postId, reason)` with `onlyOwner` protection for legal takedowns or content evading algorithmic detection.

G. Wallet-Based Authentication and Onboarding

Socio3 uses EIP-1193 wallet connectivity: `eth_requestAccounts` triggers MetaMask permission, the returned address becomes persistent identity, and React Context propagates state. For gas-free onboarding, wallets below 0.001 ETH receive 0.005 Sepolia ETH via a faucet API, with Firebase preventing duplicate claims.

H. Component Communication

Fig. 5 shows the sequence diagram for post creation, illustrating interaction between user, frontend, MetaMask, IPFS, smart contracts, and Firebase.

Three communication paths operate: blockchain writes (User → MetaMask → Ethers.js → Infura → Sepolia → Contract → Event → UI), media uploads (Image → Compression → Pinata → CID → On-chain Anchor), and real-time features (Action → Firebase SDK → Firestore → WebSocket → UI Refresh).

IV. IMPLEMENTATION DETAILS

A. Development Environment

TABLE III summarizes the environment. Contracts were compiled, tested, and deployed via Hardhat. The frontend was deployed to Vercel with GitHub CI.

B. Smart Contract Deployment

TABLE IV lists deployed addresses, verifiable via Sepolia Etherscan. Deployment used Hardhat Ignition modules for deterministic addresses.

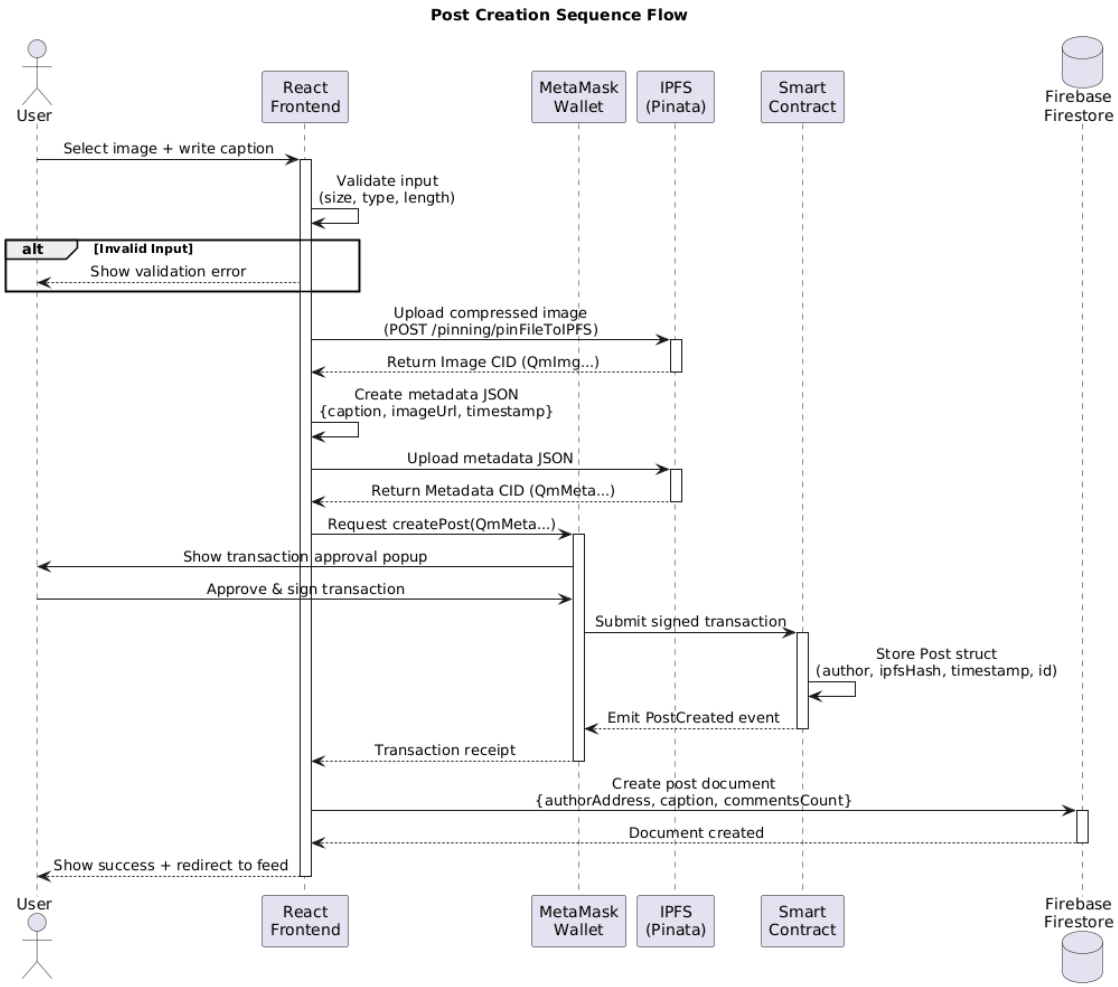


Fig. 5. Sequence diagram for post creation showing interactions across User, Frontend, MetaMask, IPFS/Pinata, Smart Contracts, and Firebase. Solid arrows are synchronous calls; dashed arrows are asynchronous responses.

TABLE III
IMPLEMENTATION ENVIRONMENT

Component	Specification
Operating System	Windows 11 / Ubuntu 24.04 LTS
Runtime	Node.js v18.0+ / npm v9.0+
Frontend Framework	React 19.1, Vite 7.1
Styling	Tailwind CSS 4.1, Framer Motion 12
Web3 Library	Ethers.js v6.15
Smart Contract Lang.	Solidity ^0.8.28
Dev Framework	Hardhat 2.26.3
Database	Firebase Firestore (v12.4)
IPFS Provider	Pinata Cloud
Blockchain	Ethereum Sepolia Testnet
RPC Provider	Infura
Wallet	MetaMask Extension
Hosting	Vercel (frontend), Sepolia (contracts)

TABLE IV
DEPLOYED CONTRACT ADDRESSES (SEPOLIA)

Contract	Address
PostContract	`0x5d5C1d313f58...37A661CF`
SocialContract	`0xedb788eb4c9D...FF57788`
ProfileContract	`0x314FBc86715e...CF903Df`

C. Frontend Architecture

The React application uses modular separation: pages (Home, Explore, Search, Profile, Upload, Post, Wallet, Admin, Messages), reusable components (Navbar, PostCard, TipModal, CommentModal), service modules abstracting blockchain/Firebase calls, custom hooks for stateful logic, and a Web3Context for global wallet state.

D. Platform User Interface

The deployed platform at `socio3.vercel.app` provides an Instagram-like social experience through a responsive dark-themed interface. Fig. 6 shows the home feed displaying posts from followed users with story bar and suggestions panel. Fig. 7 shows the user profile page with follower statistics,

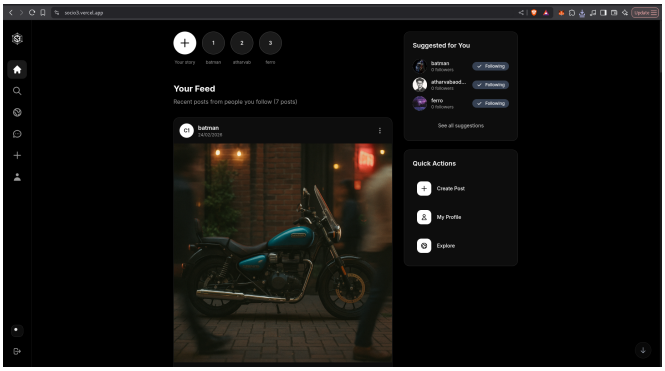


Fig. 6. Home feed page showing the story bar, personalized feed with posts from followed users, right-panel suggestions, and quick action shortcuts.

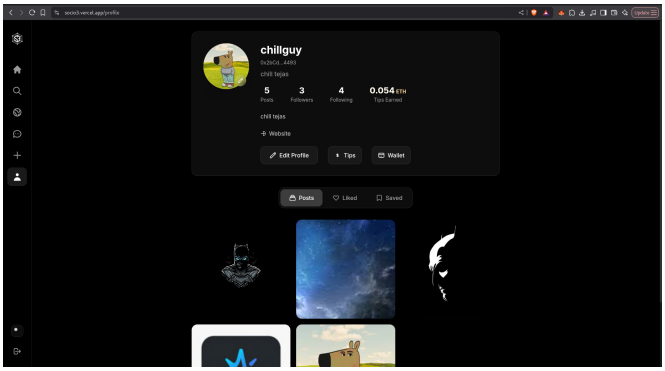


Fig. 7. User profile page displaying username, truncated wallet address, post/follower/following counts, cumulative tips earned (0.054 ETH), and a three-column post grid with IPFS-hosted media.

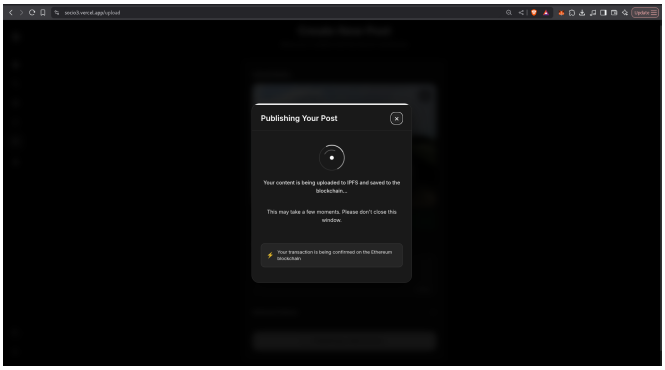


Fig. 8. Post publishing dialog at `/upload` showing the two-phase progress: IPFS content upload followed by Ethereum transaction confirmation.

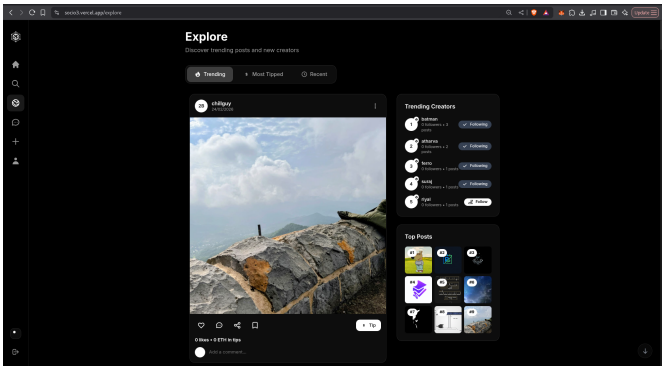


Fig. 9. Explore page at `/explore` with three filter tabs (Trending, Most Tipped, Recent), a "Trending Creators" leaderboard, and a "Top Posts" thumbnail grid.

post grid, and tip earnings. Fig. 8 shows the post publishing dialog during IPFS upload and blockchain confirmation. Fig. 9 shows the content discovery page with trending creators and top posts.

Fig. 10 shows the wallet management page where users can view their ETH balance, transaction history, and tip statistics. Fig. 11 shows Etherscan verification of on-chain transactions, demonstrating the auditability of all platform operations.

V. Results and Discussion

This section evaluates Socio3 across five dimensions: functional correctness, gas consumption, moderation effectiveness, latency characteristics, and comparative positioning.

A. Functional Testing

Twelve core operations were tested on Sepolia with Etherscan verification. TABLE V reports results—all passed with correct state transitions and event emissions.

TABLE V
Functional Test Results for Core Operations

Operation	Contract	Status
Create Profile	ProfileContract	Pass
Update Profile	ProfileContract	Pass
Change Username	ProfileContract	Pass
Create Post	PostContract	Pass
Like / Unlike Post	SocialContract	Pass
Follow / Unfollow User	SocialContract	Pass
Tip Creator (ETH)	SocialContract	Pass
Report Post	PostContract	Pass
Auto-Remove (threshold)	PostContract	Pass
Auto-Remove (low eng.)	PostContract	Pass
Auto-Remove (ratio)	Social + Post	Pass
Manual Remove (admin)	PostContract	Pass

B. Gas Consumption Analysis

TABLE VI presents gas consumption estimates, and Fig. 12 visualizes the comparison.

Profile creation consumes the most gas due to writing three storage mappings simultaneously. The three-contract architecture ensures users never load irrelevant state; a like invokes only SocialContract. At 30 gwei mainnet gas price, a like (∼55 000 gas) costs ∼$0.05 and a post (∼100 000 gas) ∼$0.09—within ranges identified as acceptable for social micro-interactions [8].

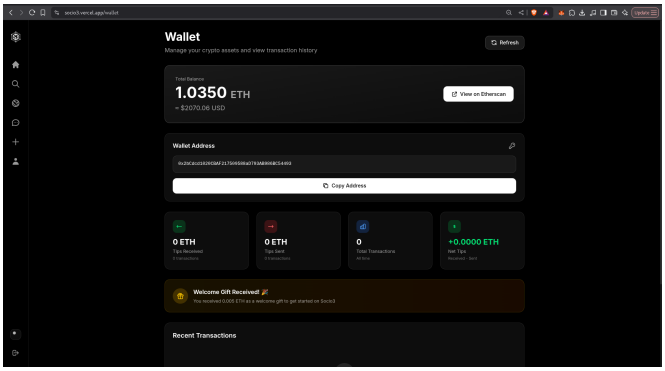


Fig. 10. Wallet management page at `/wallet` displaying total Sepolia ETH balance (1.0350 ETH ≈ $2070), wallet address, tip statistics, and the automated welcome gift notification (0.005 ETH).

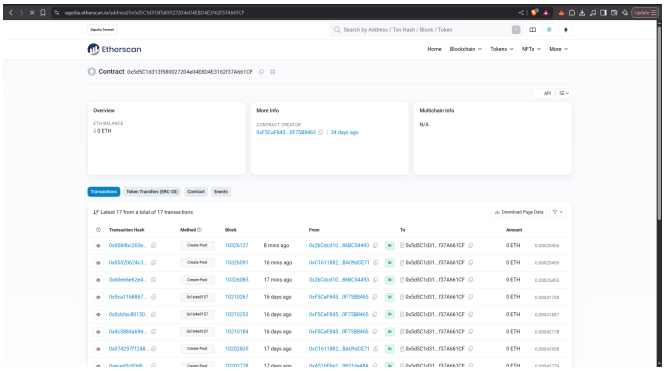


Fig. 11. Sepolia Etherscan page for the deployed PostContract at `0x5d5C...61CF`, showing 17 verified transactions with "Create Post" method calls, block numbers, and gas costs (0.00025–0.00042 ETH).

TABLE VI
ESTIMATED GAS CONSUMPTION PER OPERATION

Operation	Gas (units)	Contract
createProfile	150 000–200 000	Profile
updateProfile	45 000–65 000	Profile
createPost	80 000–120 000	Post
likePost	45 000–65 000	Social
unlikePost	30 000–45 000	Social
followUser	60 000–80 000	Social
unfollowUser	35 000–50 000	Social
tipPost	50 000–70 000	Social
reportPost	85 000–110 000	Post
manualRemovePost	50 000–70 000	Post

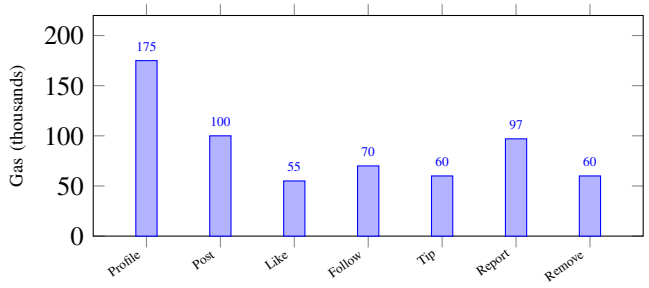


Fig. 12. Average gas consumption per operation. Profile creation is most expensive (three storage writes). Social interactions are most gas-efficient.

TABLE VII
CONTENT MODERATION SCENARIO OUTCOMES (T=5, ρ=2)

Sc.	r	l	Trigger	Removed?
1	5	10	Threshold ($r \geq T$)	Yes
2	3	0	Low engagement	Yes
3	20	10	Ratio ($20 \geq 10 \times 2$)	Yes
4	2	5	Below all criteria	No
5	4	3	Below threshold	No
6	—	—	Admin manual	Yes

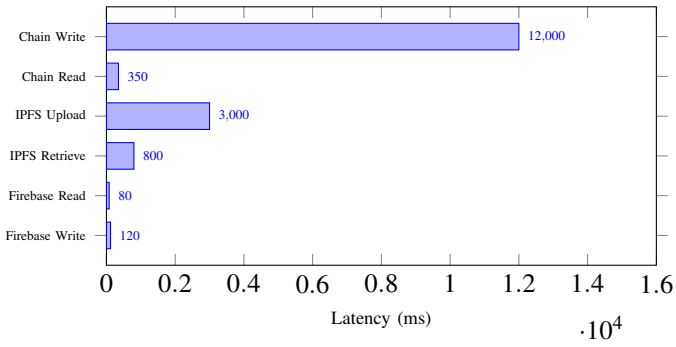


Fig. 13. Latency comparison across storage layers. Blockchain writes require ∼12 s for block confirmation; Firebase operates under 200 ms.

C. Content Moderation Effectiveness

TABLE VII presents six tested scenarios verifying the auto-removal algorithm.

All scenarios produced expected outcomes. The configurable T and ρ allow communities to tune sensitivity—addressing the concern by Al-Jaroodi and Mohamed [5] that most systems lack adaptable thresholds.

D. Latency Analysis

Fig. 13 compares response times across the three storage layers.

Blockchain writes take 12–15 s on Sepolia. Firebase operates in 50–200 ms. The platform mitigates write latency through optimistic UI: interactions reflect immediately before confirmation, with rollback on revert.

E. Comparative Analysis

TABLE VIII positions Socio3 against existing platforms.

Socio3 uniquely combines on-chain moderation with configurable thresholds, zero-fee tipping, IPFS media storage, and a consumer-ready SPA. Compared to centralized Instagram, Socio3 trades write latency for content ownership, transparent moderation, and direct creator payment.

F. Security Analysis

Four attack vectors are addressed. **Reentrancy**: mitigated via Checks-Effects-Interactions in `tipPost()` (Listing 2).

TABLE VIII
COMPARATIVE EVALUATION OF SOCIO3 AGAINST EXISTING PLATFORMS

Criterion	Socio3	Lens Protocol	Farcaster	DeSo	Steemit	Instagram
Content ownership	On-chain (user)	On-chain (NFT)	Hub network	On-chain	On-chain	Platform-owned
Moderation	On-chain algo.	Open Actions	Comm. hubs	Validator vote	DPoS vote	Centralized AI
Monetization	Zero-fee ETH tips	NFT sales	Tips + subs	Native token	Token rewards	Ad rev. share
Identity	MetaMask wallet	Profile NFT	Ethereum DID	DeSo ID	Account	Email/phone
Media storage	IPFS (Pinata)	IPFS	Off-chain hubs	On-chain	On-chain	Centralized CDN
Production UI	React SPA (✓)	SDK only	Client apps	Built-in	Built-in	Built-in
Gas-free onboard	Auto ETH gift	—	Gasless social	—	Free accounts	N/A

Access control: `onlyOwner` modifier on admin functions. **Content tampering**: IPFS CIDs are deterministic—modification produces mismatching hashes. **ETH gift exploitation**: Firebase tracking prevents cross-device duplicate claims.

G. Limitations

The `getAllPosts()` function has $O(n)$ complexity, hitting gas limits beyond $\sim$10 000 posts. The platform operates on Sepolia (no real ETH value). Pinata is a centralized IPFS dependency. Firebase comments lack blockchain immutability. Client-side search does not scale to large user bases. Username front-running is theoretically possible but negligible on testnet.

VI. CONCLUSION AND FUTURE WORK

This paper presented Socio3, a decentralized social media platform addressing centralized content ownership, opaque moderation, and extractive creator monetization through a hybrid Ethereum-IPFS-Firebase architecture. The modular three-contract design isolates security surfaces and keeps gas costs between 45 000–200 000 units per operation. The on-chain moderation algorithm with configurable thresholds (T and ρ) enables transparent, community-adaptable content governance. Functional testing across twelve operations confirmed correct execution, and comparative analysis showed Socio3 uniquely combines on-chain moderation, zero-fee tipping, IPFS storage, and a production-ready interface.

Five extensions are planned. First, Layer-2 migration (Polygon/Optimism) would reduce gas costs by $\sim$100×. Second, ERC-721 NFT minting would enable post tokenization with programmable royalties. Third, DAO governance would allow community voting on moderation parameters. Fourth, The Graph Protocol integration would replace Firebase for fully decentralized indexing. Fifth, ERC-4337 account abstraction would enable gasless transactions, removing wallet-management friction for mainstream adoption.

ACKNOWLEDGMENT

The authors thank Prof. Chaitanya Bhosale, Assistant Professor, Department of Computer Engineering, MIT Academy of Engineering, Alandi, Pune, for consistent guidance and mentorship. The authors also acknowledge the Department of Computer Engineering and MIT Academy of Engineering for providing the necessary infrastructure and resources.

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